

SEMESTER END EXAMINATIONS – MARCH 2024

Program	: B.E. – Computer Science and Engineering	Semester	: III
Course Name	: Digital Design and Computer Organization	Max. Marks	: 100
Course Code	: CS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

- Simplify the following function using K-Map, obtain the minimum POS solution and realize the circuit using NOR gates only. $F(A,B,C,D) = \prod M(0, 1, 3, 6, 8, 9, 12, 14, 15) + d(2,13)$. CO1 (08)
 - Implement the function $F(A,B,C) = \sum m(0,1,2,5,6,7)$ using CO1 (06)
 - 8:1 Multiplexer
 - 4:1 Multiplexer
using the appropriate method.
 - Briefly explain 3Bit Binary to Gray Code Converter. CO1 (06)
- Determine the Essential Prime Implicant using Tabular Method for the given function. $F(A,B,C) = \sum m(0,2,3,4,6)$. CO1 (08)
 - Simplify the following function using K-Map, obtain the minimum SOP solution and realize the circuit using NAND gates only. $F(A,B,C,D)=\sum m(0, 2, 6, 8, 10, 11, 13, 14,15)$ CO1 (08)
 - Briefly explain Priority Encoder. CO1 (04)

UNIT – II

- Draw the circuit for gated D latch using NAND gates. Describe the characteristic table, and draw the timing diagram for their operations. CO2 (08)
 - How 4 bit shift register circuit work? Show its operations with sample input sequences. CO2 (08)
 - Differentiate between asynchronous and synchronous counter circuits. CO2 (04)
- Write about JK flip flop along with its circuit, characteristic table and timing diagram. CO2 (08)
 - Analyze the design of random counter with appropriate diagrams. CO2 (04)
 - Show the working principle of three bit Asynchronous Down counter, along with an appropriate timing diagram. CO2 (08)

UNIT – III

- Write a program segment for adding n numbers using a loop, and explain in detail the execution of this program with a neat diagram. CO3 (08)
 - Convert the decimal numbers -17 and -20 into 7 bit signed 2's complement binary numbers. Perform Addition and Subtraction operations on the same numbers. Also, state whether overflow occurs or not. CO3 (06)
 - Show how the operations $X = A*(B+C) / (C-D*E)$ can be implemented in a single Accumulator based computer using i) Three address instruction ii) two address instruction iii) One-address instruction. CO3 (06)

6. a) With a neat diagram explain functional unit of a computer. CO3 (06)
b) Explain the following: i) Processor Clock ii) Basic Performance Equations iii) Performance Measurement. CO3 (08)
c) With a neat diagram with example explain Big-endian and Little-endian assignment. CO3 (06)

UNIT – IV

7. a) Perform multiplication of numbers (-12) and (-11) using booth's algorithm. CO4 (08)
b) Draw and explain the single-bus organization of the data path inside a Processor. CO4 (08)
c) Write a note on Data hazards. CO4 (04)
8. a) Multiply +13 and -6 using bit-pair recording method. CO4 (05)
b) Write and explain the control sequences for the execution of following instruction: Add (R3), R1. CO4 (05)
c) With a neat diagram, explain the floating point addition/subtraction unit. CO4 (10)

UNIT- V

9. a) Write a note on cache memory. CO5 (06)
b) With a neat diagram explain the Associative mapping functions in cache memory. CO5 (08)
c) What is an Interrupt? Explain how Simultaneous Interrupt requests are handled in computer. CO5 (06)
10. a) Illustrate the hardware required to connect an I/o device to the bus with neat diagram and also explain registers in keyboard and display Interface. CO5 (10)
b) Discuss enabling and disabling interrupts. Also explain how to handle multiple devices. CO5 (10)
